

SIMATS ENGINEERING

ASSESSMENT TOOL - II : DESIGN TASK

Course Code	ECA14
Course Name	Embedded Systems
Course Outcome	CO1
Assessment Tool	Design Task
Question No.	15, 16
Question Title	15, 16. Analyze the security vulnerabilities of a CAN bus network in an industrial plant where a penetration test demonstrated: unauthenticated message injection via the OBD-II port, bus and flooding causing compute network silence within 300ms, successful replay attacks reopening process valves. Examine the CAN protocol's inherent security limitations, evaluate available countermeasures and design a defense-in-depth security architecture without replacing the existing bus hardware
Maximum Marks	100
Bloom's Taxonomy Level	BTL 5

Evaluation Rubric (100 Marks)

Criteria	Weightage	Level 4	Level 3	Level 2	Level 1	Marks Awarded
Understanding of Design	20	☒ 20	☐ 16	☐ 12	☐ 8	20
Evaluation of Design	25	☐ 25	☒ 20	☒ 15	☐ 10	25
Justification	20	☐ 20	☒ 16	☒ 12	☐ 8	16
Suggestions for Improvement	20	☐ 20	☒ 16	☒ 12	☐ 8	16
Written Communication	15	☐ 15	☒ 12	☒ 9	☐ 6	12
Total	100					89/100

Code of Conduct:

I KEERTHIPATI HEMA SREE C192411156 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Hema

Analyze the security vulnerabilities of a CAN Bus Network

Introduction :

The Controller Area Network (CAN) bus is a reliable communication protocol widely used in industrial automation, automotive systems, and embedded control systems. However, the original CAN protocol was designed for reliability and real-time performance rather than cybersecurity. As a result, it is vulnerable to several attacks.

1. Security Vulnerabilities Identified

a) Unauthenticated Message Injection (via OBD-II Port)

- The attacker connects to the OBD-II port and injects fake CAN messages
- Since CAN has no authentication, all devices accept the forged messages

b) Bus Flooding Attack

- The attacker continuously transmits high-priority CAN frames
- Lower-priority messages are blocked due to CAN arbitration
- In the penetration test, the network became silent within 300 ms

c) Replay Attack

- The attacker records valid CAN messages
- Previously captured messages are retransmitted later
- Example: Recorded valve-open command

2. Inherent Security Limitations of CAN protocol

Limitation	Description
No Authentication	Cannot verify sender identity
No Encryption	Data is transmitted in plaintext
Broadcast communication	Every node received all messages
No Freshness check	Replay attacks are possible
Priority-based Arbitration	Enables bus flooding attacks
Small Data Field	Classical CAN supports only 8-byte payloads

3. Evaluation of countermeasures

A. HMAC-Based Message

Working

- Sender computes an HMAC using shared secret key

Advantages

- prevents message spoofing
- Ensures message integrity

Limitations

- Additional computation
- Extra bandwidth required
- Key management complexity

B. Acceptance Filtering

Working

- CAN controllers accept only predefined message IDs
- unknown IDs are discarded

Advantages

- Blocks unauthorized IDs
- Reduces attack surface

Limitations

- cannot stop replay attacks
- cannot detect compromised legitimate nodes

C. Gateway -Based Intrusion Detection System

Working

- A gateway monitors CAN traffic

Detects :

- Abnormal traffic rate
- Unknown IDs

- Replay attacks

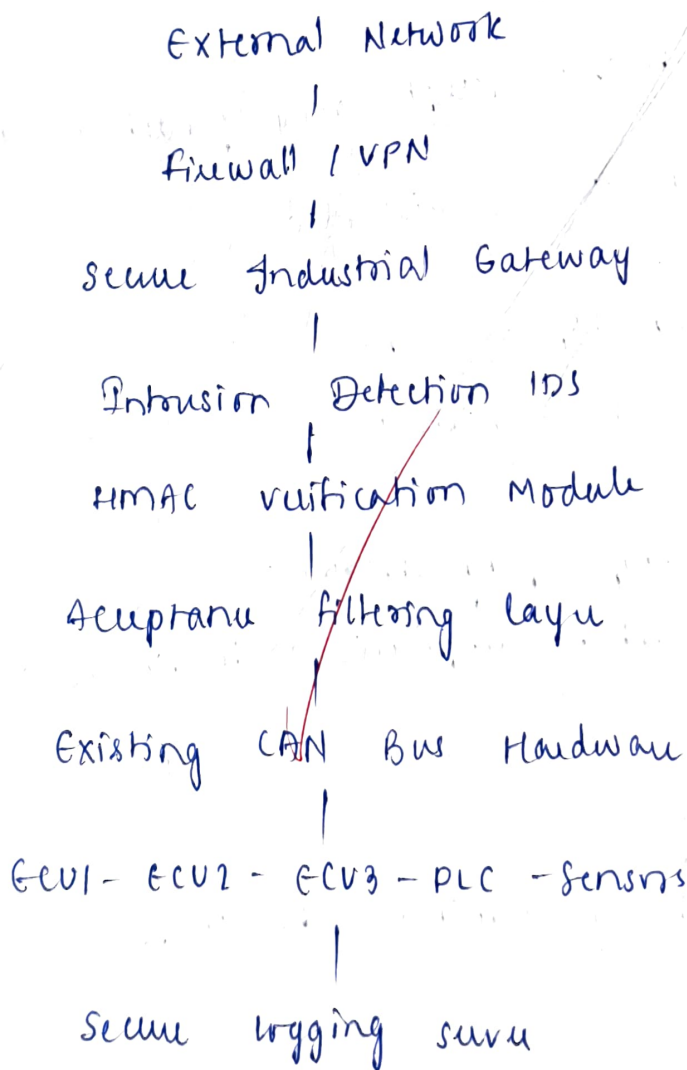
Advantages

- Detects attacks in real time
- No modification to CAN hardware

Limitation

- False positives are possible
- Requires regular rule updates

4. Defence -in- Depth Security Architecture



5. Defence layers

Layer 1 - physical security

- Restricts access to the OBD-II port
- Lock maintenance interference
- Tamper detection

Layer 2 - Network security

- Firewall and secure gateway
- Network segmentation
- VPN for remote maintenance

Layer 3 - CAN protection

- HMAC authentication
- Acceptance filtering

Layer 4 - ~~Monitoring~~

- Gateway-based IDS
- Traffic anomaly detection

Layer 5 - Response

- Block malicious nodes
- Notify operators
- Restrict normal communication automatically

G. Result

The proposed defence-in-depth architecture:

- prevents unauthorized message injection using HMAC
- Reduces spoofing through acceptance filtering
- preserves the existing CAN bus hardware while significantly improving security, reliability and safety in the industrial plant

Conclusion

The penetration test highlights that the CAN protocol lacks built-in security features such as authentication, encryption, and replay

protection. A layered security approach combining HMAC-based message authentication acceptance filtering, gateway-based intrusion detection provides effective protection

CAN FD Network Architecture

Requirements:

- Radar : 64 bytes/frame @ 50Hz
- Camera : 256 bytes/frame @ 30fps
- LiDAR : 512 bytes/frame @ 20Hz
- Ultrasonic : 8 sensors x 8 bytes/frame @ 40Hz
- Fusion ECU deadline : 20 ms
- CAN FD : 5 Mbps data phase; 64-byte payload

Proposed Architecture

- Central fusion ECU receives data from all sensors
- Radar → CAN FD BUS 1
- Ultrasonic sensors → CAN FD BUS 1
- Camera → Dedicated CAN FD Bus or Automotive Ethernet
- LiDAR → Dedicated CAN FD Bus or Automotive Ethernet

Priority Assignment

1. Radar - Highest priority
2. LiDAR - High priority

3 camera - medium priority

4. Ultrasonic - lowest priority

Bus load analysis

- Radar needs 1 CAN FD frame
- camera requires 4 CAN FD frames
- LiDAR requires 8 CAN FD frames
- Ultrasonic sensors generate multiple 'small' frames

Analysis

- CAN FDs 64-byte payload reduces protocol overhead compared to classical CAN
- 5 mbps data phase provides faster transmission
- However simultaneous transmissions from all sensors can ~~cause~~ bus contention and delay.

• During peak traffic a single CAN FD bus may not guarantee the required 20ms deadline

Recommended solution

- Use multiple dedicated sensor buses:
 - CAN FD BUS 1: Radar + ultrasonic
 - Dedicated CAN FD
- connect all buses to fusion ECU

Advantages:

- Low bus utilization
- Reduced transmission latency
- Guaranteed $< 20ms$ delivery
- Better scalability for additional ADAS sensors
- Improved reliability and fault isolation

Conclusion

A single CAN FD bus is not architecture sufficient for a level 3 autonomous driving system because camera and lidar provides very high data traffic, a multi-bus architecture with dedicated sensor buses is preferred design to ensure timely reliable sensor fusion.